Experiment: 8,9,10

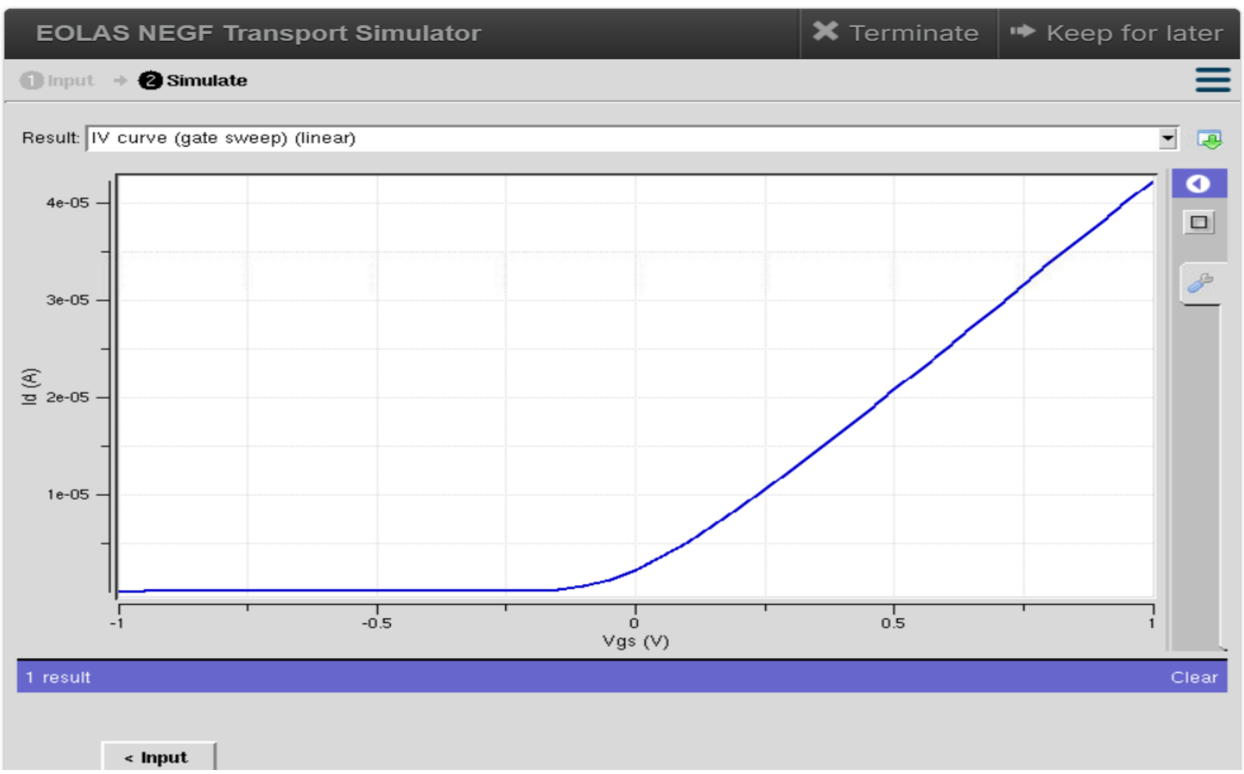
G. Harika Devi (192412592)

ECA3101-Nano Electronics

EXP 8 :Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

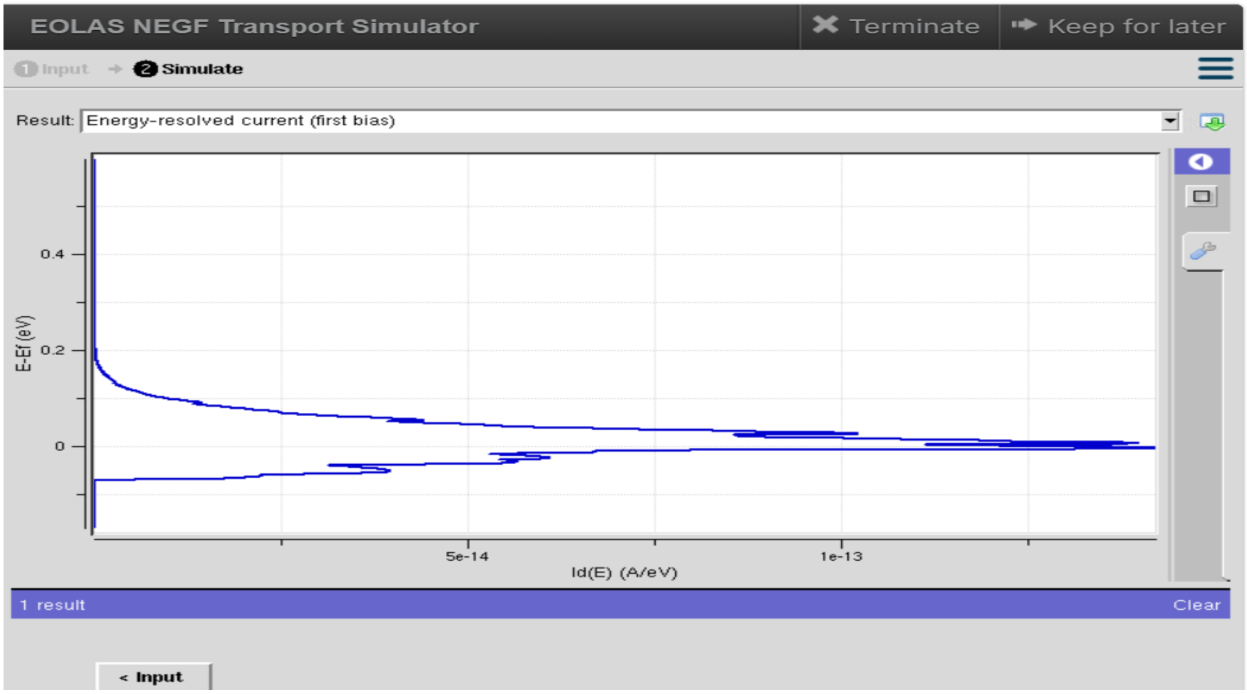
Aim:

To simulate and analyze the Voltage–Current (V–I) characteristics of FINFET, CNTFET, and GFET using NanoHUB simulation tools, and compare their electrical performance in terms of drain current, threshold voltage, transconductance, and switching characteristics.



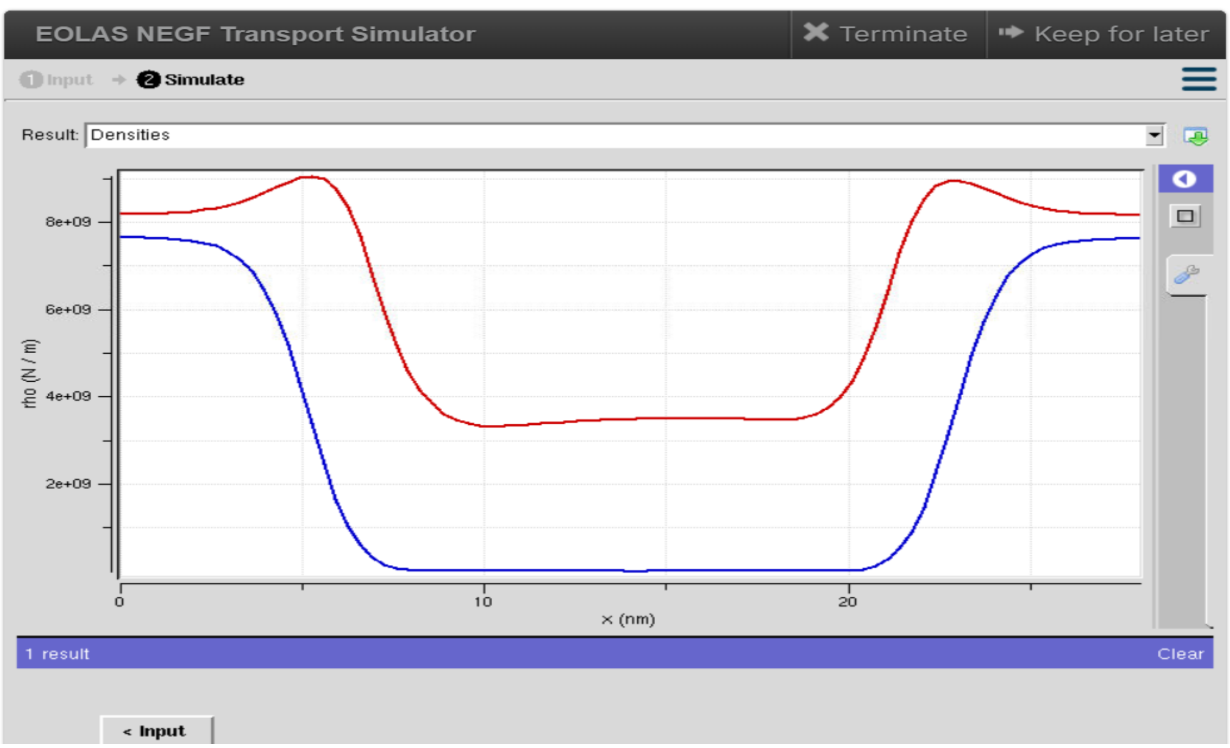
Finfet output is the Transfer Characteristics (Id–Vgs) of a transistor. It shows how the drain current (Id) changes with the gate-to-source voltage (Vgs) while keeping the drain voltage (Vds) constant.

TEST CASE 1: Energy Resolved currents (first bias)



Graph shows how much current is carried by electrons at different energy levels under the applied bias voltage.

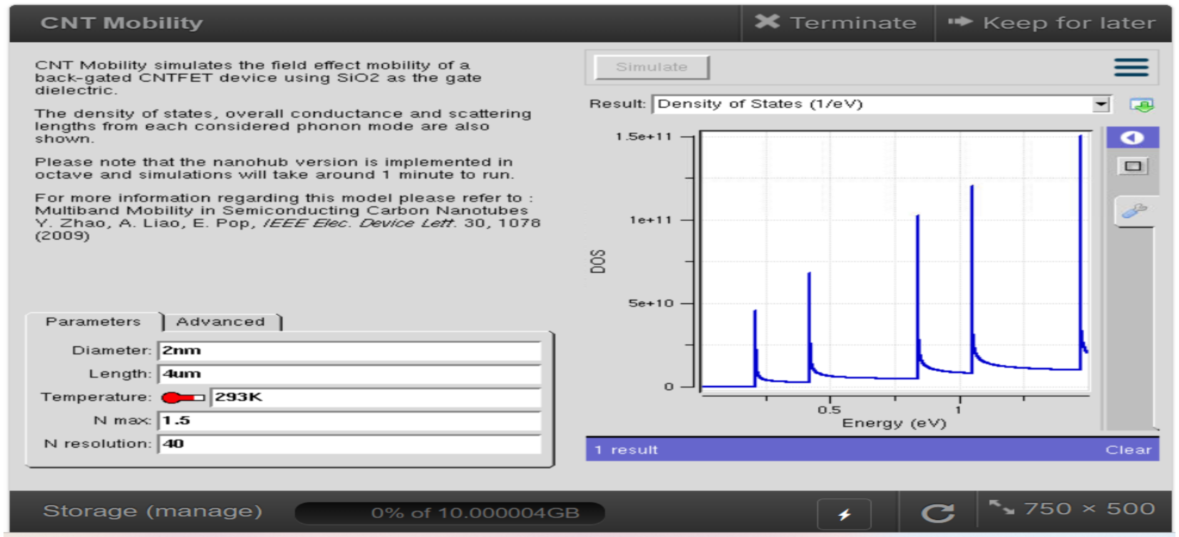
TEST CASE 2: Densities



- **Red Curve:** Electron density

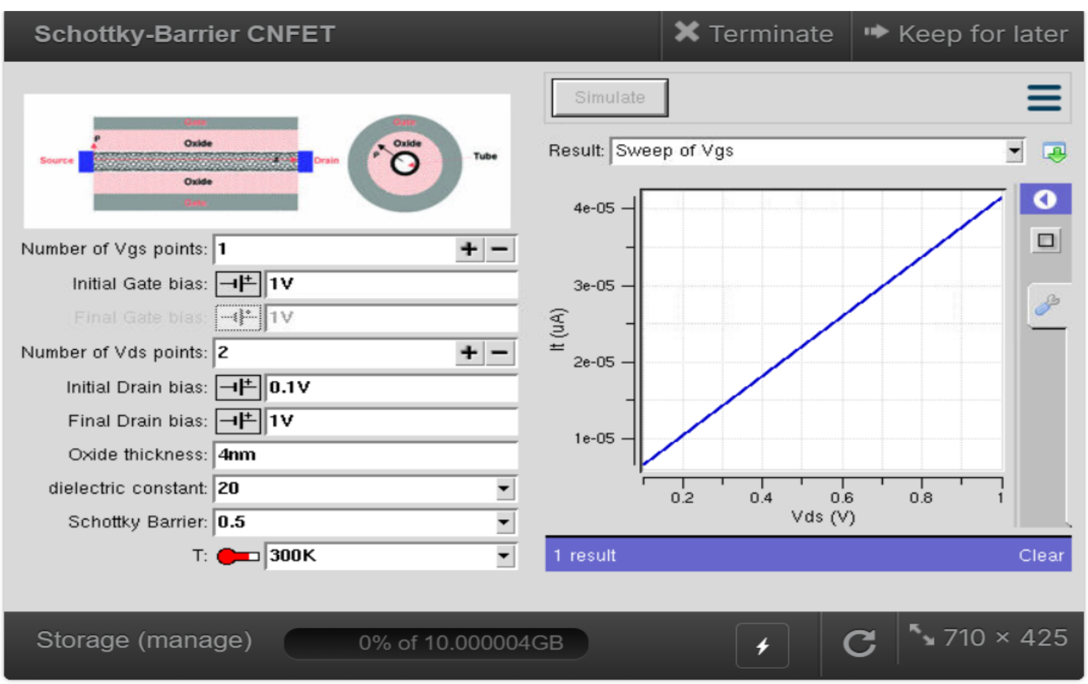
- **Blue Curve:** Hole density (or another carrier density distribution depending on the simulator settings)

CNTFET



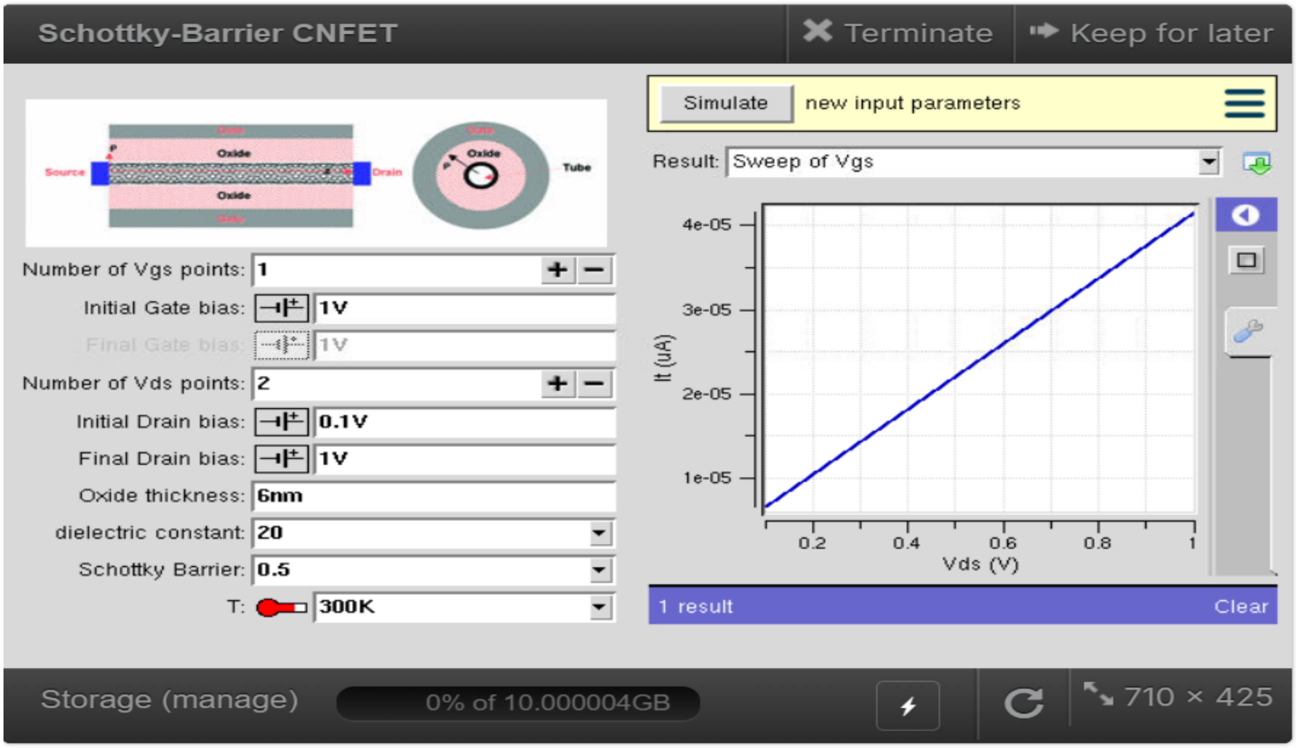
They show the Density of States (DOS) vs Energy graph obtained from the CNT Mobility tool in NanoHUB for a Carbon Nanotube Field Effect Transistor (CNTFET**).**

TEST CASE 1: thickness=4nm



The Output Characteristics (Drain Current I_D vs Drain Voltage V_{DS}) of a Schottky Barrier Carbon Nanotube Field Effect Transistor (SB-CNFET).

TEST CASE 2:Thickness=6nm



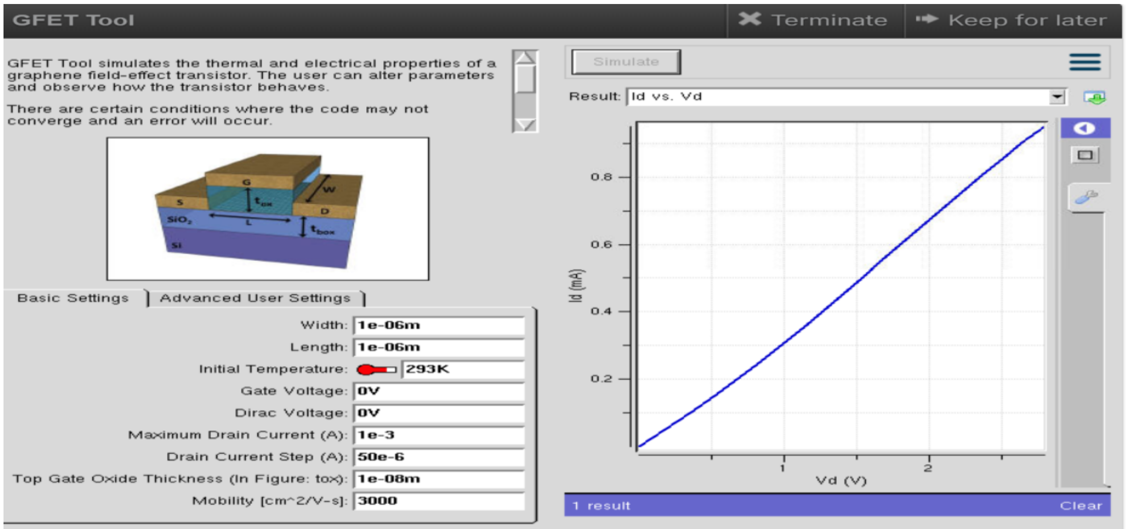
The graph shows that the drain current increases linearly with increasing drain voltage. This indicates efficient carrier transport through the carbon nanotube channel and stable operation of the Schottky-Barrier CNTFET. The high current confirms the excellent conductivity of carbon nanotubes.

GFET: TEST CASE 1: MAXIMUM DRAIN CURRENT =1xe^-3

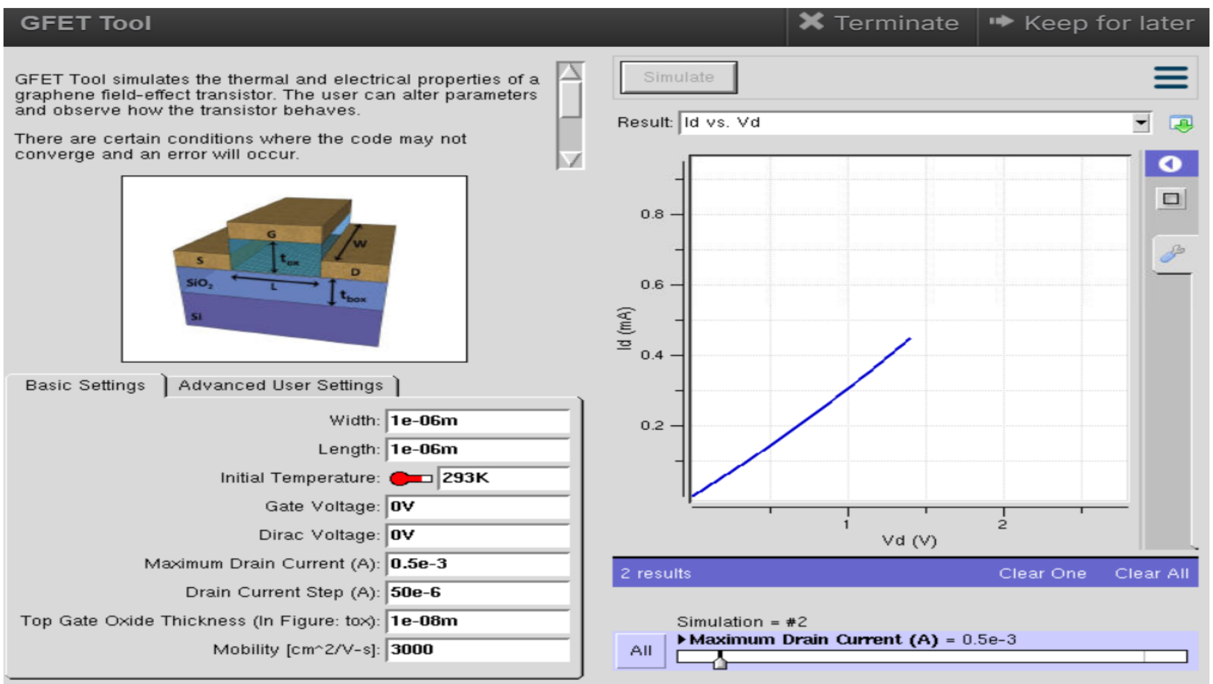
- X-axis: Drain Voltage (V_D) in volts.
- Y-axis: Drain Current (I_D) 3in milliamperes (mA).
- As the drain voltage increases from 0 V to approximately 3 V, the drain current increases almost linearly until it reaches nearly 1 mA.

Impact

- The graph shows that graphene provides very high carrier mobility, allowing electrons to move quickly through the channel.
- Increasing the drain voltage increases the electric field, resulting in a proportional increase in drain current.



TEST CASE:2 MAXIMUM DRAIN CURRENT =0.5 e^-3



- The graph demonstrates that the GFET provides stable and efficient current conduction even with a lower maximum drain current limit.
- Setting the maximum drain current to 0.5 mA limits the highest current allowed during the simulation while maintaining the same operating behavior.

- **Conclusion**
- The V–I characteristics of FINFET, CNTFET, and GFET were successfully simulated using NanoHUB and their electrical performances were compared. The results show that FINFET provides excellent electrostatic control with low leakage current, CNTFET offers higher carrier mobility and faster switching with lower power

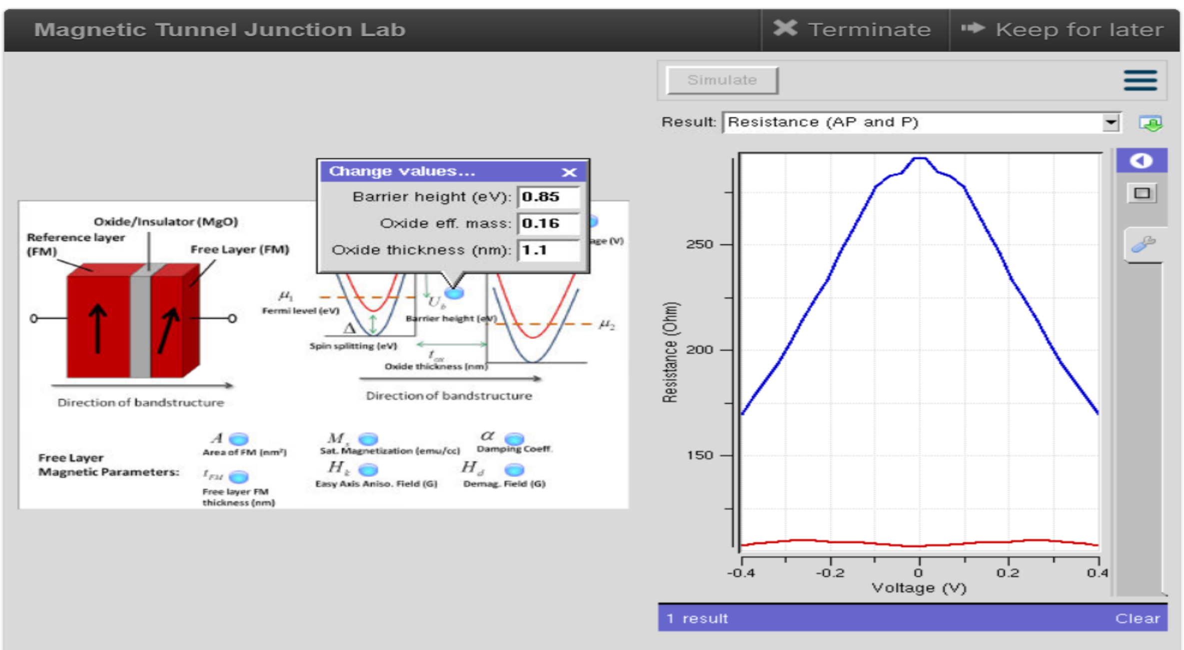
consumption, and GFET exhibits the highest carrier mobility and current conduction, making it suitable for high-frequency applications. Overall, CNTFET and GFET outperform conventional silicon-based devices in terms of speed and efficiency, while FINFET remains an effective technology for low-power integrated circuits.

EXP 9: Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

Aim

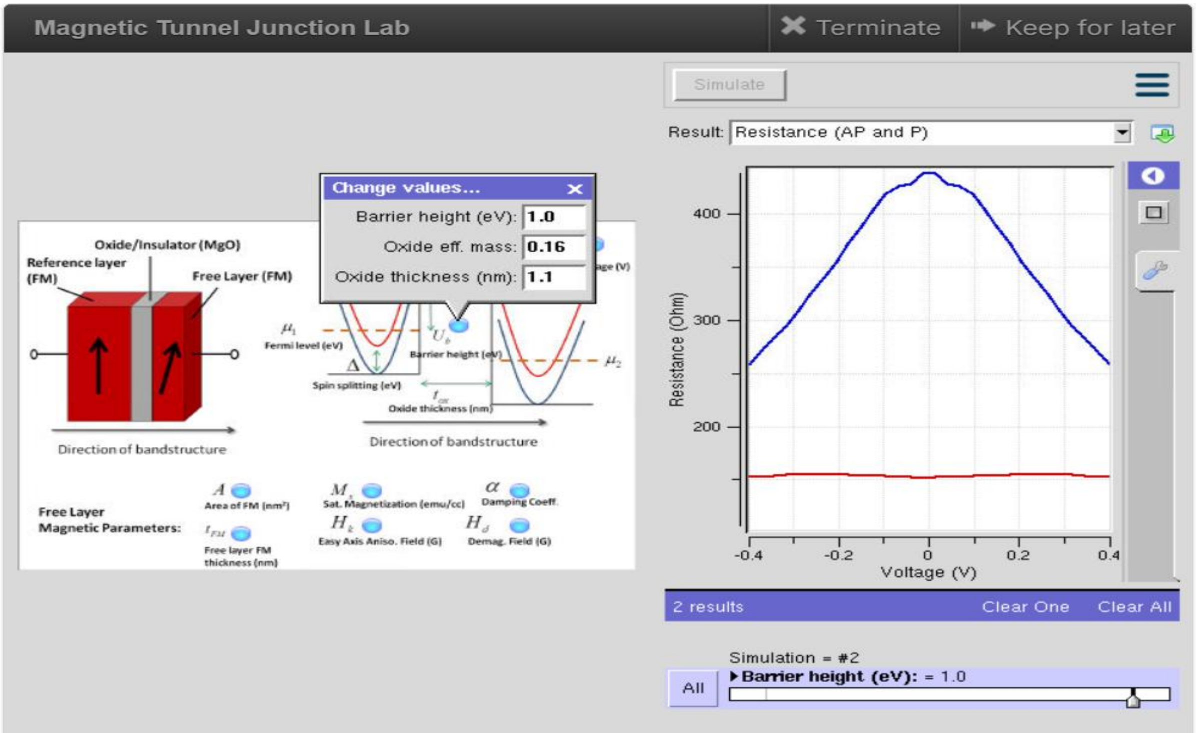
To simulate the Tunneling Magnetoresistance (TMR) characteristics of a Magnetic Tunnel Junction (MTJ) by varying the barrier height using NanoHUB and to analyze its effect on the resistance in Parallel (P) and Anti-Parallel (AP) magnetic configurations.

TEST CASE 1: Barrier Height=0.85eV



- X-axis: Applied Voltage (V)
- Y-axis: Resistance (Ω)
- The blue curve represents the Anti-Parallel (AP) resistance, while the red curve represents the Parallel (P) resistance.

TEST CASE 2: Barrier Height = 1.0eV



- X-axis: Applied Voltage (V)
- Y-axis: Resistance (Ω)
- Increasing the barrier height to 1.0 eV increases the tunneling resistance because electrons encounter a higher energy barrier.

Impact:

- **X-axis:** Applied Voltage (V)
- **Y-axis:** Resistance (Ω)
- Increasing the barrier height to **1.0 eV** increases the tunneling resistance because electrons encounter a higher energy barrier.

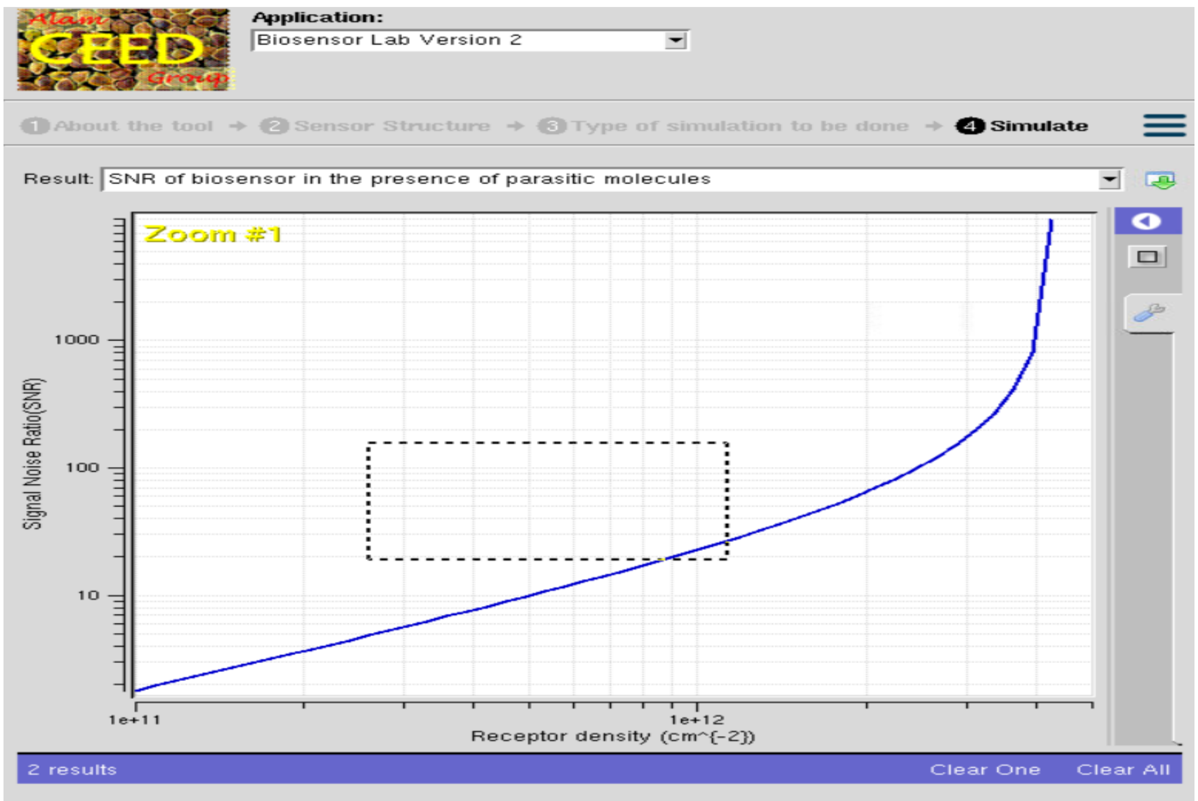
Conclusion:

The NanoHUB simulation demonstrates that the barrier height strongly influences the Tunneling Magnetoresistance (TMR) characteristics of a Magnetic Tunnel Junction. At 0.85 eV, the MTJ exhibits lower resistance and efficient spin-dependent tunneling with a high TMR ratio, while increasing the barrier height to 1.0 eV increases the tunnel resistance and reduces tunneling current, improving insulation and data retention. Therefore, selecting an appropriate barrier height is essential to achieve an optimal balance between high TMR, low power consumption, fast switching, and reliable operation in spintronic devices such as MRAM and magnetic sensors.

EXP 10: Biosensor and Bio-FET characteristics using NanoHub

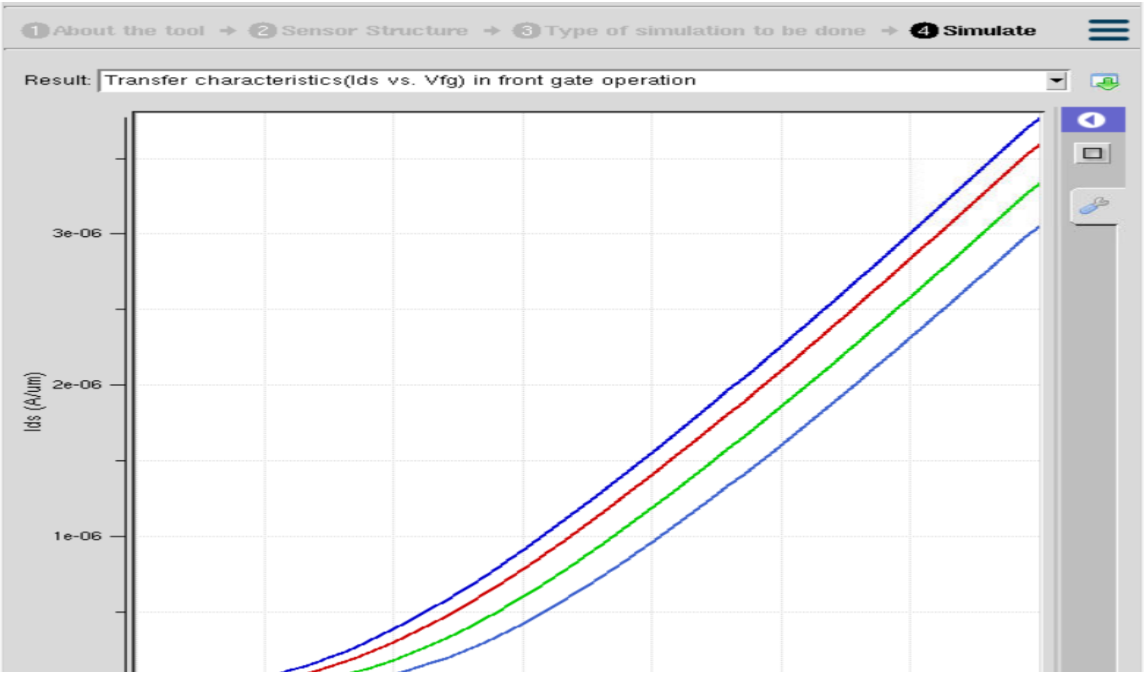
Aim

To simulate the characteristics of a Biosensor and Bio-FET (Biological Field Effect Transistor) using NanoHUB and analyze the effect of biomolecule charge and concentration on the electrical characteristics of the Bio-FET.



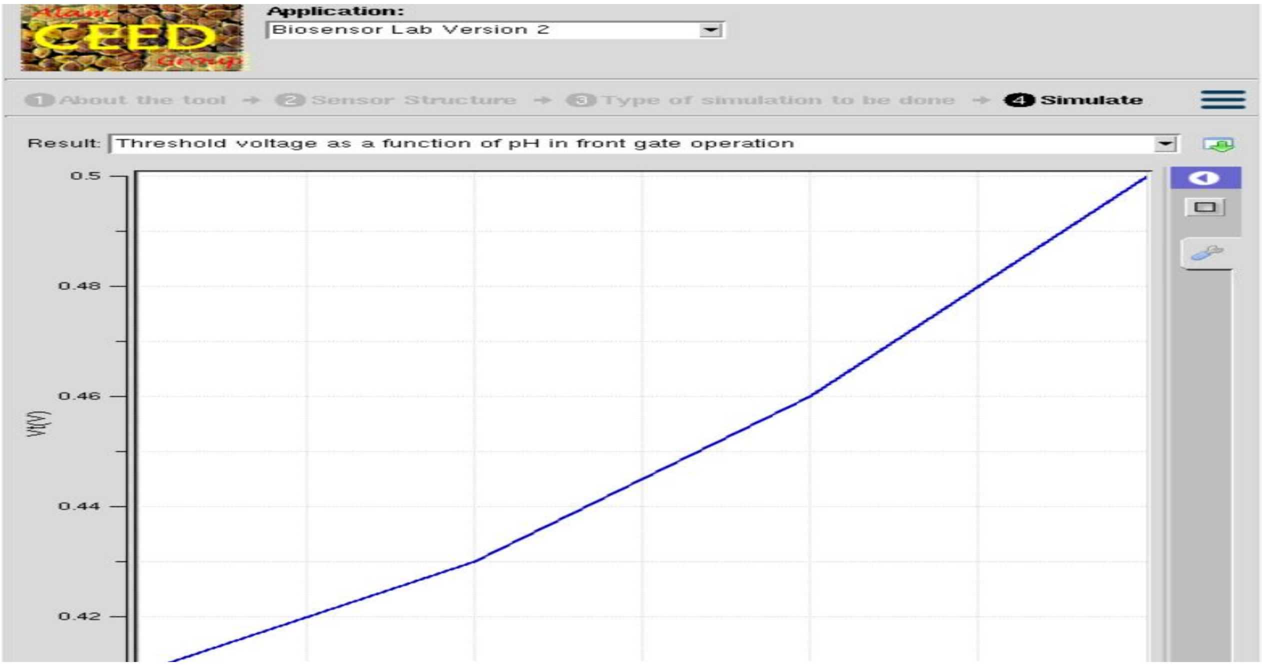
- X-axis: Receptor Density (cm^{-2})
- Y-axis: Signal-to-Noise Ratio (SNR)
- The graph shows that the Signal-to-Noise Ratio (SNR) increases as the receptor density increases.
- At low receptor densities, the SNR is small because fewer biomolecules are captured by the sensor.

TEST CASE 1: Transfer Characteristics (Idg vs Vfg) front gate



- X-axis: Front Gate Voltage (Vfg)
- Y-axis: Drain Current (Ids)
- The graph shows that the drain current (Ids) increases as the front gate voltage (Vfg) increases.
- The different colored curves represent different sensing conditions (such as different biomolecule concentrations or charges**).**

TEST CASE2: Threshold voltage as a function of Ph in front gate operation



The graph shows that the threshold voltage increases with increasing pH. This occurs because the surface potential at the gate changes due to hydrogen ion concentration. The linear variation indicates good pH sensitivity of the biosensor**.**

Impact

Bio-FETs enable rapid, label-free, highly sensitive, and real-time detection of biological molecules, making them ideal for medical diagnostics and environmental monitoring

Conclusion

The Bio-FET simulation demonstrates that the electrical characteristics of the device are significantly influenced by the presence of charged biomolecules near the sensing surface. Changes in biomolecule concentration alter the surface potential, resulting in variations in drain current and threshold voltage. These characteristics make Bio-FETs highly sensitive, fast, and suitable for real-time biosensing applications such as disease diagnosis, DNA detection, glucose monitoring, and environmental sensing